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Feature-Engineering Enabled Multi-Objective Evolutionary Impedance Fitting Technique

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***Abstract*—The impedance fitting technique aims to make the impedance of an equivalent circuit the same as the measured impedance of the real system through parameter tuning. Since only the magnitude and phase of the impedance are measurable, the single-objective optimization algorithm, where the objective is to minimize the weighted summation of the fitting errors on magnitude and phase, has been widely used to achieve automatic impedance fitting. However, this scheme is easy to be trapped in the local minima, which, leads to poor fitting precision. A feature-engineering enabled multi-objective evolutionary impedance fitting (FEMEIF) technique is proposed in this paper. By implementing several effective feature engineering techniques to fully leverage the limited impedance information, and by utilizing these synergetic features to enable multi-objective impedance fitting for realizing better hierarchy on individuals, FEMEIF much improves the impedance fitting performance with higher fitting precision and lower optimization variance compared with the traditional single-objective impedance fitting schemes. FEMEIF provides a novel methodology for improving the impedance fitting task, requires no additional measurements or expenses, and is universal for various optimization algorithms or equivalent circuits. Besides, it provides an open topic for exploring other potential features to achieve further improvement. The effectiveness of the FEMEIF is verified separately on fitting the impedance of a generic RLC-parallel circuit and on an industrial application of fitting the common-mode impedance of a motor system.**

***Index Terms*—Impedance fitting, feature engineering, multi-objective evolutionary algorithm.**

I. Introduction

Impedance fitting is one of the most critical industrial processes for various industrial applications which require the use of a specific equivalent circuit to represent the real physical systems, such application scenarios include identifying the parameters of a lithium battery for online health monitoring [1]-[2], using the equivalent circuit for the electromagnetic interference (EMI) design [3]-[5], applying the equivalent circuit into the time-domain power system simulations [6], etc. Generally, the equivalent circuit is constructed by a specific combination of resistors, capacitors, and inductors with initially unknown values. Through careful parameter tuning on components, the impedance of equivalent circuit can be identical to the real physical system within a specific frequency span. In this way, the equivalent circuit can behave a similar external performance as the real physical system and could surrogate it for many industrial applications.

The weighted summation of the mean squared fitting error of the magnitude and phase (also named *WSMSE*) is the only widely-recognized criterion for evaluating the fitting precision in industries because it directly quantifies the mismatch between the fitted and the target impedance. Therefore, the impedance fitting task is basically a single-objective optimization problem, it aims to find appropriate components' value to minimize *WSMSE*. To avoid inefficient human-labor-based parameter tuning, various optimization algorithms have been proposed to automatically identify the components' value of the equivalent circuit. Among them, gradient algorithms such as Gauss-Newton algorithm, Levenberg-Marquardt algorithm have successfully been applied in commercial impedance fitting software on account of the fast convergence and light calculation [7]-[9]. However, gradient algorithms are very easy to be trapped into the local minima, which results in an over-reliance on a good initial parameter [10]. To deal with this problem, swarm evolutionary algorithms such as genetic optimization, particle swarm optimization, differential evolutionary optimization, etc., are proposed [10]-[16]. These algorithms are initialized with a group of parameter candidates and could provide adequate "mutate" mechanisms for escaping from the local minima, which, allows a significant reduction in the fitting error. However, due to the biomimetic and model-free properties, the evolutionary algorithm usually shows undesirably unstable fitting results and slow convergence speed. Based on that, some white-box optimizing methods, such as Bayesian optimization and Tree-structured parzen estimator, can also be applied for the impedance fitting tasks to ensure the comprehensibility and stability of the optimization process [17]. These methods utilize a surrogate model to online learn the solution space, and then reasonably recommend parameter candidates instead of random evolution. However, the calculation consumption of updating the surrogate model rapidly increases with the historical data, which, makes it too expensive and inefficient to be applied in impedance fitting applications.

As can be seen, although evolutionary algorithms have the potential to be an ideal way for dealing with impedance fitting tasks owing to a good balance between fitting precision and calculation cost, it still suffers from unsatisfactory fitting precision and unstable fitting performance due to the over-random evolution. Thus, the evolution-based impedance fitting task is still facing huge challenges. Except for the improvements in algorithms, improving the equivalent circuit to ease the fitting process [3], [18], or increasing the

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measurement to provide more information for impedance fitting [2] are also effective approaches to help improve the fitting performance. However, these methods are usually expensive and show poor generalization. As can be seen, previous researchers paid too much concentration on studying complicated algorithms, equivalent circuits, or measurement techniques, but rarely noticed that improving the way of constructing the optimization objectives can easily break the bottleneck that the evolutionary algorithm has met. According to the survey, nearly all previous works only used the *WSMSE* as the optimization objective because it is the most widely-used evaluation indicator for fitting precision [10]-[16]. In fact, the impedance information is far less than fully leveraged. Treating the features of the impedance as the optimization objectives can help the evolution of *WSMSE* escape from local minima, and achieve better fitting precision.

This paper proposes a feature-engineering enabled multi-objective evolutionary impedance fitting (FEMEIF) technique to solve the single-objective impedance fitting task in a multi-objective way. By mining several effective features from the limited impedance information to enable multi-objective optimization, FEMEIF can realize a better hierarchy of the individuals during the optimization process under the common effects of multiple feature-based objectives. In result, FEMEIF achieves better fitting precision and lower optimization variance compared with the traditional single-objective impedance fitting technique without introducing additional measurements or expenses. The main contributions are:

1) In previous studies, the design of feature engineering techniques for impedance fitting tasks has rarely been studied, and the efficient way of applying feature engineering techniques for improving impedance fitting performance remains undiscovered. This paper proposes several novel feature engineering techniques to process the limited impedance information into feature-based auxiliary optimization objectives. These feature-based objectives can retain "shortcut" individuals omitted by *WSMSE* to help the optimization process of *WSMSE* escape from local minima.

2) Instead of improving algorithms, circuits, or measurements, FEMEIF enhances the impedance fitting performance from a new perspective (optimization objectives). FEMEIF breaks through the limitation of the previous related works in which only single-objective (*WSMSE*) impedance fitting has been used. More importantly, FEMEIF verifies that solving a single-objective optimization task in a multi-objective way (based on feature engineering techniques that have "shortcut" property) can achieve much better performance than solving it in the traditional single-objective way. This new concept can also be generalized for other industrial single-objective optimization applications.

In this study. Sec. II introduces the feature engineering techniques, Sec. III introduces the framework of FEMEIF. Sec. IV validates the effectiveness of the FEMEIF through experiments on two practical applications. Sec. V gives the conclusion and the future directions.

II. Feature Engineering For Impedance Fitting

Feature engineering is a process of using expert knowledge to manipulate and transform the raw data into features to fully expose its characteristics, properties, etc. It is widely used in machine learning domains to leverage the limited data and enhance model accuracy. In fact, feature engineering can also be applied to optimization algorithms as the optimization objectives. In this section, one widely used feature and another five newly proposed features would be introduced. It is worth noting that the frequencies mentioned below are in $\log_{10}$-form.

A. Mean squared Error

Mean squared error (*MSE*) is the most widely used criterion for evaluating the fitting precision in nearly all the previous works. The *MSE* in impedance fitting tasks is formulated as

$$MSE = \sum_{i=1}^{M} (Z_{(i)}^{*} - Z_{(i)}^{fit})^2 / N_M \quad (1)$$

where N_M denotes the total number of the sampled frequency points, $Z_{(i)}^{*}$ and $Z_{(i)}^{fit}$ separately denote the target and the fitted impedance at the i-th frequency. The penalty given by *MSE* increases with the fitting error in a square relationship. Usually, in nearly all the previous impedance fitting works, the *MSE* of the magnitude and the phase of impedance Z are separately calculated and then summed up by a weight coefficient α to form the final optimization objective, which is also named weighted sum *MSE* (*WSMSE*) [9]-[16]. The final fitness score F_{WSMSE} is obtained by calculating the reciprocal of *WSMSE* as

$$F_{WSMSE} = N_M / \sum_{i=1}^{M} ((Z_{mag(i)}^{*} - Z_{mag(i)}^{fit})^2 + \alpha (Z_{pha(i)}^{*} - Z_{pha(i)}^{fit})^2) \quad (2)$$

B. Shape similarity

Only using *WSMSE* for evaluating the fitting performance loses much useful information and sometimes may even mislead the optimization route. Fig. 1 shows an example, as Individual 2 has a lower Euclidean distance to the target impedance than Individual 1 over the whole frequency domain, Individual 2 would have a higher ranking if only *WSMSE* is used. However, in fact, Individual 1 should be better than

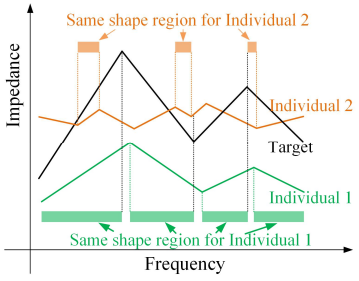


Fig. 1. Example for illustrating the effectiveness of *SS*.

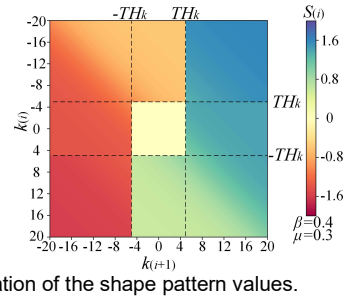


Fig. 2. Calculation of the shape pattern values.

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TABLE I
THE CALCULATION OF THE SHAPE PATTERN VALUE

$S_{(i)}$	$k_{(i+1)}<-TH_k$	$-TH_k<k_{(i+1)}<TH_k$	$k_{(i+1)}>TH_k$
$k_{(i)}<-TH_k$	$Fx(-1, k_{(i+1)}, k_{(i)})$	$Fx(-1, k_{(i+1)}, k_{(i)})$	$Fy(1, k_{(i+1)}, k_{(i)})$
$-TH_k<k_{(i)}<TH_k$		0	$Fx(1, k_{(i+1)}, k_{(i)})$
$k_{(i)}>TH_k$	$Fy(-1, k_{(i+1)}, k_{(i)})$	$Fx(1, k_{(i+1)}, k_{(i)})$	

Individual 2 because Individual 1 has a much more similar shape to the target impedance, Individual 1 can well fit the target impedance only by simply giving some bias whereas the equivalent circuit in Individual 2 requires large amounts of parameter tuning.

Shape similarity (*SS*) is a criterion that only evaluates the similarity of the trends between two impedance curves regardless of the Euclidean distance. In impedance fitting applications, the calculation of the *SS* mainly contains 3 steps:

Step 1: Calculate the slope of the impedance at each frequency point $f_{(i)}$ by

$$k_{(i)} = (Z_{(i+1)} - Z_{(i)}) / (f_{(i+1)} - f_{(i)}) \quad (3)$$

Step 2: Calculate the shape pattern values $S_{(i)}$ for each frequency point by Table I, where, TH_k denotes a threshold for $k_{(i)}$, β ($0<\beta<0.5$) and μ denote hyperparameters which determine the $S_{(i)}$, $Fx(\cdot)$ and $Fy(\cdot)$ are calculated by (4) and (5). The relationship between $S_{(i)}$ and the impedance slope is visualized in Fig. 2. As can be seen, $S_{(i)}$ basically describes the trends of the impedance curve. The sign symbol of $S_{(i)}$ describes whether the impedance is rising ($S_{(i)}>0$), keeping ($S_{(i)}=0$), or falling ($S_{(i)}<0$). The absolute value $|S_{(i)}|$ describes whether the current impedance trend is slowing down ($1-\beta<|S_{(i)}|<1$), steady ($|S_{(i)}|=1$), speeding up ($1<|S_{(i)}|< 1+\beta$), or trend reversal (it means the trend turns from rising to falling or from falling to rising, $1+\beta<|S_{(i)}|<1+1.5\beta$). The larger difference between $|S_{(i)}|$ and 1 indicates a greater acceleration.

$$Fx(x, k_{(i+1)}, k_{(i)}) = x(\beta(1 - 2/(1 + e^{\mu x(k_{(i+1)} - k_{(i)})})) + 1) \quad (4)$$

$$Fy(y, k_{(i+1)}, k_{(i)}) = y(3\beta/2(1 + e^{-\mu y(k_{(i+1)} - k_{(i)} - 2yTH_k)}) + 1) \quad (5)$$

Step 3: Calculate the final fitness score of *SS* between the predicted impedance Z^{fit} and the target impedance Z^* by (6). Obviously, if an individual has more identical impedance trend in a wider frequency domain as the target impedance, the fitness score of *SS* would be higher.

$$F_{SS} = 1/(\sum_{i=1}^{M} | S_{(i)}^* - S_{(i)}^{fit} | / N_M) \quad (6)$$

C. Peak and trough consistency

In the impedance curves, the peaks and troughs are caused by the resonance of the distributed inductors and the distributed capacitors. Thus, the frequencies of the peaks and troughs in the target impedance can directly reflect the real values of inductors and capacitors. In practical impedance fitting tasks where human-based tuning is used, the relative location of the peaks/troughs between the fitted impedance and the target impedance is a critical criterion for guiding the tuning of the inductance and capacitance in the equivalent circuit. The individuals whose peak/trough frequencies are more identical to the target impedance usually indicate a better estimation precision on the inductance and capacitance, therefore, these individuals are supposed to get higher ranking because they can well fit the target by minor tuning on capacitance and inductance in the equivalent circuit.

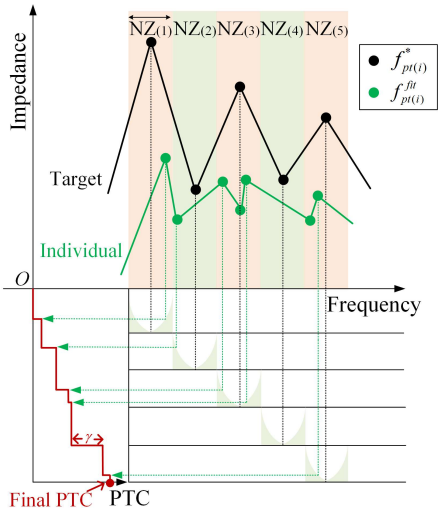


Fig. 3. Calculation of *PTC*.

Based on that, peak and trough consistency (*PTC*) is another effective feature to be extracted. *PTC* aims to evaluate the degree of consistency between the peaks/troughs of the target impedance and the peaks/troughs of the fitted impedance. There are mainly two steps for calculating *PTC*.

Step 1: As shown in the purple and blue blocks in Fig. 3, divide the whole frequency domain into several neighbor-zones $NZ_{(i)}$ (i = 1, 2, …, N_z) for each peak/trough $f^*_{pt(i)}$ in the target impedance Z^*. It is recommended to set the central frequency of the adjacent peak and trough as the boundary of each $NZ_{(i)}$.

Step 2: Individually visit $f^*_{pt(i)}$. for each $f^{fit}_{pt(j)}$ within $NZ_{(i)}$ ($f^{fit}_{pt(j)}$ is the j-th peak/through in the fitted impedance Z^{fit}):

If $f^{fit}_{pt(j)}$ has the same attribute as $f^*_{pt(i)}$ ($f^*_{pt(i)}$ and $f^{fit}_{pt(j)}$ are both peaks or they are both troughs), then calculate a cost score and accumulate it into the *PTC* as:

$$PTC += (e^{\gamma|f^*_{pt(i)} - f^{fit}_{pt(j)}|/(f_{\max} - f_{\min})} - 1) \quad (7)$$

where $f_{\max}$ and $f_{\min}$ denote the upper and lower bound of the whole frequency domain, γ denotes a hyperparameter which determines the utmost penalty. If $f^{fit}_{pt(j)}$ has the opposite attribute from $f^*_{pt(i)}$ (e. g. $f^*_{pt(i)}$ is peak(or trough) but $f^{fit}_{pt(j)}$ is trough(or peak)), no penalty is added because introducing this penalty may give preference for those "lazy" individuals (number of $f^{fit}_{pt(j)}$ are far less than the number of $f^*_{pt(i)}$) than those individuals who have similar numbers of predicted peaks (and troughs) as the target impedance but make too many mistakes on such peak-trough mismatch. As a result, introducing this kind of penalty usually worsens the optimization result. Besides, if no $f^{fit}_{pt(j)}$ exists within the neighbor-zone of $f^*_{pt(i)}$ ($f^{fit}_{pt(j)}$ and $f^*_{pt(i)}$ are in the same attribute), a max penalty would be given for this fitting omission by $PTC += \gamma$. Finally, the final fitness score F_{PTC} can be obtained by calculating the reciprocal of *PTC*.

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Algorithm1: Calculation of *DTW* score

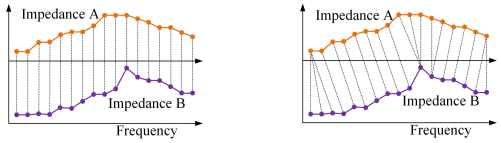
Input: A, B: two impedance sequences sized N_A, N_B, separately; D: distance matrix
Output: *DTW*

```
for i = 1 to N_A
    for j = 1 to N_B
        M_C[i, j] = min(M_C[i-1, j], M_C[i, j-1], M_C[i-1, j-1]) + D[i, j]
    end for
end for
LastRoute = (N_A, N_B)
Route.append(LastRoute)
DTW = D[LastRoute.x, LastRoute.y]
repeat
    LastRoute = argmin(M_C[LastRoute.x - 1, LastRoute.y - 1],
                       M_C[LastRoute.x - 1, LastRoute.y],
                       M_C[LastRoute.x, LastRoute.y - 1])
    Route.append(LastRoute)
    DTW += D[LastRoute.x, LastRoute.y]
Until LastRoute = (1, 1)
return DTW
```

As can be seen, *PTC* automatically realizes the correlation between the target peaks/troughs and the fitted peaks/troughs. For each uniquely correlated target peak/trough and fitted peak/trough, *PTC* gives the penalty according to their frequency distance. For the target peak/trough which has no or over one corresponding fitted peaks/troughs within its neighbor-zone, *PTC* also gives reasonable penalties.

D. Dynamic time warping

Dynamic time warping (*DTW*) analysis is another criterion for evaluating the shape similarity between two impedance curves [19]. *DTW* is able to seek the minimum Euclidean distance between aligned series through shape scaling and shifting. As is exhibited in Fig. 4, if only *WSMSE* is used for evaluating the fitting performance, the two impedance curves would be justified to be significantly different because *WSMSE* is basically a Euclidean matching. In fact, the shape and magnitude of the two curves are very similar except for some irregular scaling. This scenario is also supposed to be given a high fitness score because it usually indicates good parameter estimation precision on all the components in equivalent circuit owing to the good accordance on the number of the impedance peaks/troughs and a decent Euclidean distance over the whole frequency range. Based on that, *DTW* can be used as a criterion to give a preference on this scenario. There are mainly two steps for applying *DTW* into impedance fitting tasks:



(a) Euclidean distance (b) *DTW*

Fig. 4. Difference between Euclidean distance and *DTW*.

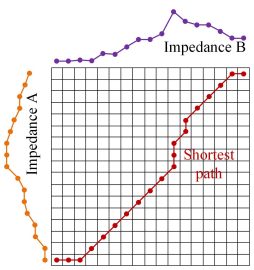


Fig. 5. Calculation of *DTW*.

Step 1: As shown in Fig. 5, construct a 2-dimention distance matrix D whose i-jth element is the absolute error between the target impedance at the i-th frequency point and the fitted impedance at the j-th frequency point.
Step 2: Find the shortest path (with least accumulated distance) through dynamic programming, and the summation of elements in this shortest path is the *DTW* distance. The calculation of *DTW* is detailed in Algorithm 1. The final fitness score F_{DTW} can be obtained by calculating the reciprocal of *DTW*.

Different from the *SS* which only concerns the trend of the impedance curve, *DTW* evaluates shape similarity in a more comprehensive view which includes Euclidean distance, scaling, shifting, etc. Therefore, *DTW* and *SS* are different but complementary. It is worth mentioning that calculating *DTW* would be time-consuming because the time complexity of the dynamic programming is $O(N_M^2)$, where N_M is the number of the sampled frequency points [19]. To solve it, downsampling technique can be used to reduce the size of the distance matrix [20]. N_D denotes the number of sampled frequency points.

E. Low-frequency priority

Expert knowledge about how to efficiently tune the parameters can also be utilized and processed into the features. Recalling the human-based impedance fitting process, it is always recommended to fit the impedance from the low-frequency domain to the high-frequency domain. This is because the low-frequency impedance is mainly determined by the major components with large values in the equivalent circuit, whereas high-frequency impedance is not only influenced by the major components but is also sensitive to many distributed components with small values. Therefore, it would be more efficient to first determine the major components by getting a good fitting precision in the low-frequency domain and then tune those minor components to realize the impedance fitting in the whole frequency domain. As is illustrated in Fig. 6, when two individuals have the same fitness score on *MSE*, the one with better fitting precision in the low-frequency domain (Individual 1) should be given a higher fitness score. There are two steps to calculate the low-frequency priority (*LFP*):
Step 1: Divide the whole frequency domain into several neighbor-zones $NZ_{(i)}$ (i = 1,…, N_z) as step 1 of the *PTC* calculation.
Step 2: From the low-frequency $NZ_{(1)}$ to the high-frequency $NZ_{(N_Z)}$, calculate the *MSE* $E_{NZ(i)}$ for the impedance of each neighbor-zone $NZ_{(i)}$. If $E_{NZ(i)}$ is less than a pre-set threshold TH_{lf}, then $F_{LFP} += 1$ until when $E_{NZ(i)}$ is larger than TH_{lf}.

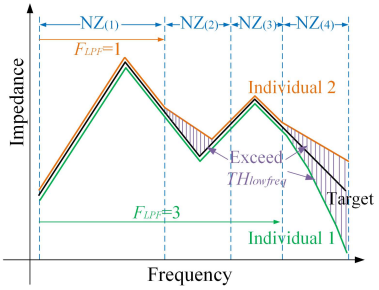


Fig. 6. Example for illustrating the effectiveness of *LFP*.

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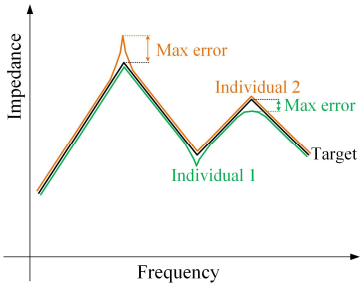


Fig. 7. Example for illustrating the effectiveness of *LMEP*.

As can be seen, *LFP* score basically counts the number of the successive well-fitted neighbor-zones from low frequency to high frequency. The wider the well-fitted successive low-frequency region is, a higher fitness score F_{LFP} would be got. Those individuals that only have good fitting precision in the high-frequency domain but show poor precision in the low-frequency domain are not preferred under *LFP*, which, can effectively guide the evolution to a better optimization route.

F. Low maximum error preference

In most impedance fitting tasks, an overlarge fitting error in some certain frequency points is usually unacceptable because it acts like the "short plank" in the "cask effect", which, generally does more harm for the practical application of the equivalent circuit than those individuals with uniform small errors in the whole frequency domain. Based on that, As is illustrated in Fig. 7, although the two individuals have the same fitness score under *MSE*, the individual 1 should be given higher fitness score due to the lower maximum absolute error. The calculation of the fitness score of the low maximum error preference (*LMEP*) can be expressed as

$$F_{LMEP} = 1/(Max([|Z^*_{(1)} - Z^{fit}_{(1)}|, \ldots, |Z^*_{(M)} - Z^{fit}_{(M)}|])) \quad (8)$$

In summary, the introduced feature engineering techniques all show preference for individuals which may have poor fitting precision F_{WSMSE}, but can easily reach a good F_{WSMSE} with only minor parameter tuning. Therefore, these individuals are named "shortcut" individuals. Processing these features into objectives can help the optimization process of *WSMSE* escape from the local minima. This is the design principle of feature engineering techniques for impedance fitting tasks.

III. Multi-objective Evolutionary Impedance Fitting

The workflow of the proposed FEMEIF technique is shown as Fig. 8. Compared with the traditional *WSMSE*-based evolutionary impedance fitting (WBEIF) technique which uses a single-objective evolutionary algorithm to optimize F_{WSMSE}, FEMEIF not only optimizes F_{WSMSE}, but also treats any combination of other features (F_{SS}, F_{PTC}, F_{DTW}, F_{LFP}, F_{LMEP})

TABLE II
Decision Space of Each Component in Two Cases

Case I		Case II	
Variable	Value range	Variable	Value range
R_1 - R_6	[1, 10^5] Ohm	R_1 - R_7	[10^{-2}, $10^{4.5}$] Ohm
C_1 - C_6	[10^{-9}, 10^{-4}] F	C_1 - C_4	[10^{-12}, 10^{-7}] F
L_1 - L_6	[10^{-7}, 10^{-2}] H	L_1 - L_4	[10^{-9}, 10^{-4}] H

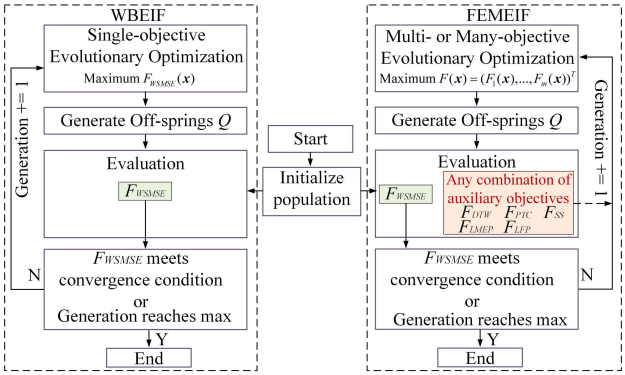


Fig. 8. Comparison of workflow of traditional WBEIF and FEMEIF.

as objectives to be simultaneously optimized. The optimization objective is formulated as:

$$\text{Maximum } F(\boldsymbol{x}) = (F_1(\boldsymbol{x}),...,F_m(\boldsymbol{x}))^T,\ m \geq 2 \quad (9)$$
$$\text{Subject to } \boldsymbol{x} \in \boldsymbol{\Omega}$$

where $\boldsymbol{x} = (x_{(1)}, x_{(2)}, \ldots, x_{(Nc)})^T$ is a N_C-dimensional solution vector (also named as an individual) from the decision space $\boldsymbol{\Omega}$ which is given in Table II. F_i (i=1, …, m) denotes the fitness score of the i-th objective. At least 2 objectives should be optimized, one should be F_{WSMSE} and the others are any combination of the proposed auxiliary objectives.

Multi- or many-objective optimization algorithm can be used to implement the FEMEIF technique. First, select an appropriate optimization algorithm, and define the objectives as well as the constraints. At each generation, a group of individuals (child individuals of the last generation) with evaluated multiple objectives (F_{WSMSE} + any combination of auxiliary objectives) are sent to the optimization algorithm. Based on fitness scores of multiple objectives, the parent individuals of the current generation are chosen by multi-/many-objective elitist strategy. Through genetic procedures (selection, crossover, and mutation), child individuals of the current generation can be generated. The detailed implementation procedures above follow the same procedure as the selected multi-/many-objective optimization algorithm. Then, for each child individual, the fitness score of each objective is evaluated. Since F_{WSMSE} (fitting precision) is the only concerned indicator for the impedance fitting task, only F_{WSMSE} is used to justify whether the optimization process has converged, this procedure is the same as the single-objective algorithm. If the converge condition hasn't been met, the evaluated child individuals would be returned back to the multi-/many-optimization algorithm for further evolution.

There are two main differences between the proposed FEMEIF with the traditional many-objective optimization tasks. First, although we enable many-objective optimization, the impedance fitting task is basically a single-objective task. Therefore, F_{WSMSE} is still used as the criterion for evaluating the fitting performance and justifying whether the optimization has converged. The other proposed objectives are named feature-based auxiliary objectives because they are only used for elitist strategy and selection process to promote the optimization of *WSMSE* through picking and maintaining "shortcut" individuals omitted by *WSMSE*. Obtaining solutions with

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TABLE III
COMPONENT VALUES IN RLC-PARALLEL GENERIC EQUIVALENT CIRCUIT

Variable	Target Value	Variable	Target Value	Variable	Target Value
R_1	2000 Ohm	R_3	500 Ohm	R_5	800 Ohm
C_1	10 uF	C_3	1 uF	C_5	0.501 uF
L_1	1585 uH	L_3	100 uH	L_5	1.259 uH
R_2	3000 Ohm	R_4	1200 Ohm	R_6	50 Ohm
C_2	3.98 uF	C_4	0.631 uF	C_6	0.158 uF
L_2	10 uH	L_4	2.512 uH	L_6	0.251 uH

TABLE IV
COMPARISON BETWEEN WBEIF AND FEMEIF (ALL OBJECTIVES APPLIED) AT 300TH GENERATION (CASE I)

Algorithm	Mean *WSMSE*	Variance of *WSMSE*	Mean HV
WBEIF	10.5	28.8	0.25
FEMEIF (NSGA-II)	5.91	10.4	0.93
FEMEIF (HypE)	5.43	10.3	1.02
FEMEIF (GrEA)	**4.85**	**10.1**	**1.15**
FEMEIF (MOEA/D)	5.02	10.9	1.10

TABLE V
HYPERPARAMETER SETTINGS OF WBEIF (#), FEMEIF (EQUIPPED WITH NSGA-II (*), GREA (^), HYPE (%), MOEA/D(&))

Variable	Description	Case I	Case II
$N_M^{\#*\wedge\%\&}$	Number of sampled frequency points	180	96
$N^{\#*\wedge\%\&}$	Population size	200	150
$\lambda^{\#*\wedge\%\&}$	Mutate rate	0.1	0.08
$\eta_m^{\#*\wedge\%\&}$	Distribution index of polynomial mutation	1	1
$\eta_c^{\#*\wedge\%\&}$	Distribution index of simulated binary crossover	1	1
$N_T^{\#*\wedge\%}$	Hyperparameter in tournament selection	2	2
$\alpha^{\#*\wedge\%\&}$	Weights between magnitude and phase	25.12	17.38
$TH_k^{*\wedge\%\&}$	Threshold for slope	8	5
$\beta^{*\wedge\%\&}$	Bound of the pattern value calculate	0.4	0.4
$\mu^{*\wedge\%\&}$	Parameter in shape pattern value calculate	0.5	0.3
$N_D^{*\wedge\%\&}$	Downsampled number of frequency points	60	48
$TH_{lf}^{*\wedge\%\&}$	Threshold in *LPF* calculation	1.6	1
$\gamma^{*\wedge\%\&}$	Penalty coefficient in *PTC* calculation	0.41	0.41
$N_{div}^{\wedge}$	Number of grid divisions	15	25
$N_N^{\&}$	Number of neighborhoods of each weight	20	-

higher comprehensive quality (higher similarity, lower maximum error) is just an extra bonus and is not critical.

Second, different from the traditional many-objective optimization where conflicting objectives are required, the impedance fitting task is basically a single-objective task although we solve it in a many-objective way. F_{WSMSE} is the only concerned criterion. Therefore, the auxiliary objectives should be weakly or moderately harmonic with *WSMSE*. On the one hand, weak harmony confirms some potential individuals which are evaluated from different aspects of *WSMSE* can be picked. On the other hand, the optimization of the auxiliary objectives can also simultaneously improve *WSMSE* owing to synergy.

It is worth noting that all the proposed auxiliary objectives are not necessarily simultaneously applied, any number of these objectives can be individually or combinedly used. Considering that the computational complexity would increase with the increasing objectives, the objective design scheme of FEMEIF will be detailed in Sec. IV.

IV. EXPERIMENTS

In this section, the fitting performance of the proposed FEMEIF would be verified in two practical cases. The first case is to fit the impedance of a generic RLC-parallel equivalent circuit, the second case is to fit the common-mode impedance of a real motor drive system. The most widely used WBEIF is used for comparison. In order to avoid fortuitous results and to conduct the probability analysis, the experiments are independently run 100 times with different stochastic processes. As *WSMSE* is the only widely-recognized criterion for evaluating the fitting precision, other objectives are just used for promoting *WSMSE* and are not suitable to be used as the final evaluation criterion (e. g. users will never really think individual 1 in Fig. 1 is better than individual 2 if they are the final results), we mainly concern the loss of *WSMSE*.

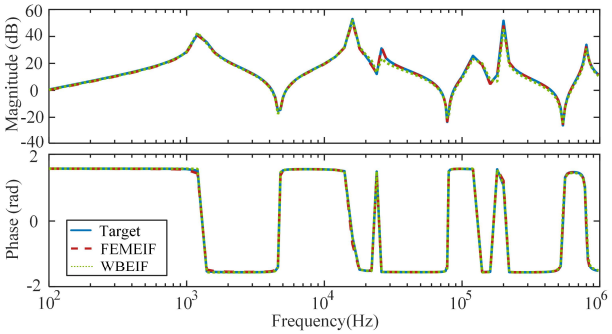


Fig. 9. Comparison of fitting results between the traditional WGEIF and the FEMEIF (all objectives) at the best run (Case I).

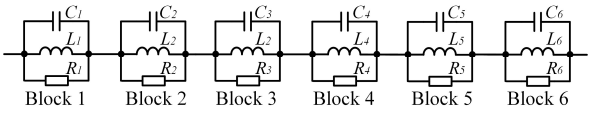


Fig. 10. Topology of the target generic equivalent circuit.

A. Case I: Generic RLC-parallel equivalent circuit

The target impedance to be fitted is visualized as the blue curve in Fig. 9, it originates from an RLC-parallel circuit that contains six in-series connected RLC-parallel blocks as shown in Fig. 10, the target value of each component is given in Table III. Such RLC-parallel topology is one of the most widely used generic equivalent circuits in practice because it theoretically can fit any given impedance as long as the length of the blocks is properly designed [13]. In this case, an RLC-parallel circuit which also contains six RLC blocks with unknown component values is used to fit the target impedance. Thus, it can theoretically realize zero-error fitting. As the selection of the equivalent circuit also influence the final fitting performance and is another independent topic, Case I aims to get rid of the influence of the equivalent circuit and purely compare the utmost fitting capability of the algorithms.

At first, the *WSMSE* at the final convergence stage (300th generation) of the WBEIF and the FEMEIF (all auxiliary objectives applied) equipped with different optimization algorithms are compared in Table IV. Although *WSMSE* is the only evaluation indicator in impedance fitting tasks, we still use Hypervolume (HV) indicator [21] to verify that FEMEIF can obtain high-quality solutions not only on *WSMSE* but also on many other aspects (better shape similarity, lower maximum fitting error, etc.). For HV calculation, fitness on each objective

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TABLE VI
SPEARMAN CORRELATION ANALYSIS BETWEEN DIFFERENT OBJECTIVES

R_s	*WSMSE*	*PTC*	*DTW*	*LFP*	*SS*	*LMEP*
WSMSE	0	0.31	0.48	0.44	0.56	0.48
PTC	0.31	0	0.39	0.42	0.13	0.15
DTW	0.48	0.39	0	0.58	0.27	0.41
LFP	0.44	0.42	0.11	0	0.51	0.36
SS	0.56	0.13	0.27	0.51	0	0.26
LMEP	0.48	0.15	0.41	0.36	0.26	0

should be normalized to 0 to 1, all the comparing algorithms share the same reference point (the value of each dimension of the reference point is set as 1.1 [22]). The selected algorithms for implementing FEMEIF are 1) NSGA-II [23]: A classical Pareto-based multi-objective optimization. 2) GrEA [24]: A typical Relaxed dominance-based many-objective optimization. 3) HypE [25]: A typical Indicator-based many-objective optimization. 4) MOEA/D (Tchebycheff) [26]: A typical Aggregation-based many-objective optimization. Single-objective genetic algorithm is used for implementing the traditional WBEIF [27]. The hyperparameters of these algorithms are given in Table V. For a fair comparison, the common parameters are set the same. Besides, the common critical operations (e. g. simulated binary crossover [28] and polynomial mutation [29]) are also set the same. It is worth noting that for fairness, the elitist strategy of WBEIF is set similarly to GrEA, which, is selecting N best (highest F_{WSMSE}) individuals from N(parents) + N(children) individuals of the current generation as the parents of the next generation.

The experimental results show that no matter which algorithm is equipped, FEMEIF not only realizes a better fitting precision (lower mean *WSMSE*) and stabler fitting performance (lower variance) than the traditional WBEIF at the final convergence stage, it can also obtain solutions with higher comprehensive qualities (higher HV). It is a reasonable result because FEMEIF simultaneously optimizes multiple objectives. NSGA-II shows the worst fitting precision among all the comparing objective-combination algorithms. An enhanced many-objective algorithm can usually achieve better fitting precision because the "shortcut" individuals picked by the auxiliary objectives can be well retained by an appropriate distribution during the optimization process, which, enables them to constantly function to help improve *WSMSE*. In the following experiments, GrEA, which achieves the best fitting performance, is used to equip with FEMEIF to conduct detailed analyses about the individual and combined effects of the proposed feature-based auxiliary objectives.

Before studying the effectiveness of the proposed objectives, the relation between *WSMSE* and the auxiliary objectives is worth analyzing. Following [30], Spearman correlation is used. The calculated Spearman correlation coefficient R_s is within -1 to 1, the sign of R_s indicates whether two objectives are harmonic (+) or conflicting (-). $|R_s|$ indicates the degree of correlation. The calculation results are listed in Table VI. It can be seen that R_s between *WSMSE* and other objectives are all within 0.31 to 0.56, which, indicates a weak to moderate harmonic relationship. This is the basis of the feasibility of the proposed auxiliary objectives as has been analyzed in Sec. III.

Fig. 11 and Fig. 12 conduct several ablation experiments to verify the individual and combined effects of the feature-based auxiliary objectives. The dashed line represents the best run. The solid line represents the mean *WSMSE* of 100 independent runs. The semitransparent region around the solid line represents the confidence interval (CI) of the mean fitting performance with a confidence level 0.95, which, indicates that we can expect 95% confidence of capturing the true mean fitting performance within this interval [31]. The calculation of the CI of the mean fitting performance with a confidence level

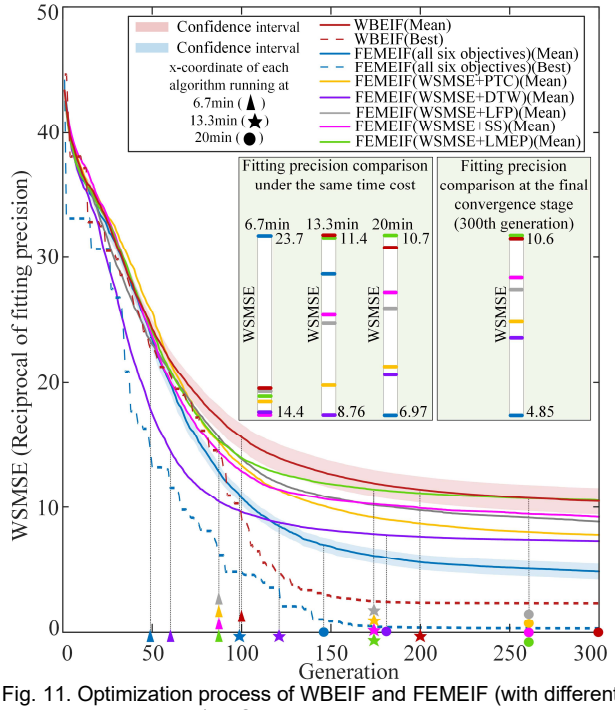


Fig. 11. Optimization process of WBEIF and FEMEIF (with different objective-combination) in Case I.

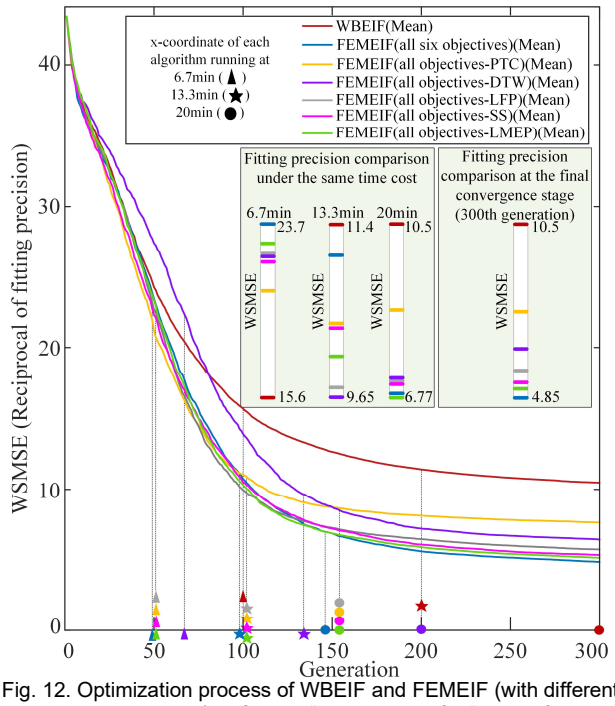


Fig. 12. Optimization process of WBEIF and FEMEIF (with different objective-combination) in Case I, "-" represents for "except for".

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TABLE VII
TIME COST OF WBEIF AND FEMEIF RUNNING 300 GENERATIONS
"-" REPRESENTS "EXCEPT FOR"

Algorithm	Time cost Case I / Case II
WBEIF	20 min /15 min
FEMEIF (*WSMSE+PTC/SS/LFP/LMEP*)	23 min /17 min
FEMEIF (*WSMSE+DTW*)	33 min /24 min
FEMEIF (*ALL objectives - DTW*)	30 min /22 min
FEMEIF (*ALL objectives –PTC/SS/LFP/LMEP*)	39 min /27 min
FEMEIF (*ALL objectives*)	41 min /29 min

α_c (=0.95 empirically) can be expressed as:

$$\overline{Y} = \sum_{i=1}^{i=N_R} Y_{(i)} / N_R \tag{10a}$$

$$S^2 = \sum_{i=1}^{i=N_R} (Y_{(i)} - \overline{Y})^2 / (N_R - 1) \tag{10b}$$

$$CI = \overline{Y} \pm Z_{\alpha_c/2} \times S / \sqrt{N_R} \tag{10c}$$

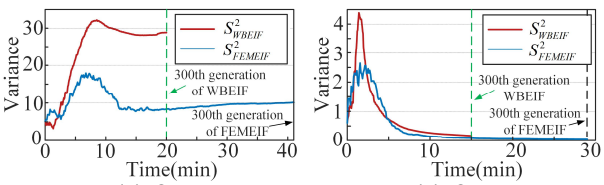
where, N_R denotes the number of samples (independent runs) which equals 100. $Y_{(i)}$ denotes the loss of the i-th run at a certain generation. $Z_{\alpha_c/2}$ is the Z-value for confidence level α_c.

Comparing the mean *WSMSE* at the final convergence stage (300th generation) in Fig. 11 and Fig. 12, it can be seen that although WBEIF concentrates on optimizing *WSMSE*, the FEMEIF with any number of auxiliary objectives still achieves better final fitting precision because the feature-based auxiliary objectives can effectively help the optimization process of *WSMSE* escape from the local minima. Different objectives show different effectiveness for improving final fitting precision, their effectiveness can be approximately ranked as *PTC*≈*DTW*>*SS*≈*LFP*>*LMEP*. *LMEP* seems didn't work when it is individually applied, but it can function well when combined with other auxiliary objectives according to Fig. 12. With the combined objectives increasing, the mean fitting precision of FEMEIF usually gets better. This is because the correlation between the auxiliary objectives is also mutually weak/moderate synergy as depicted in Table VI. Therefore, when multiple objectives are combined, each auxiliary objective can individually function to promote *WSMSE*. The combination of all six objectives achieves the best fitting precision at final convergence stage.

However, the main concern of enabling multi-objective impedance fitting is that the time cost per generation increases with the number of objectives. Table VII gives the time consumption of the traditional WBEIF and FEMEIF with different objective combinations (the fitness score of each objective is calculated in parallel). Clearly, the time cost much increases with the increasement of objectives, besides, calculating *DTW* is also time-consuming as has been analyzed in Sec. II. D. In Fig. 11 and Fig. 12, bar-subgraphs are added to compare the *WSMSE* loss of each algorithm at 6.7min, 13.3min and 20min (end time of WBEIF). It can be seen that except for the initial stage where all algorithms are far from convergence, FEMEIF (with any objective-combination) still shows a superior fitting precision than WBEIF under the same time consumption. This is because the auxiliary objectives can significantly reduce the *WSMSE* loss, which, provides surplus compensation for the influence of the increased time-consumption. In addition, it is noticed that *DTW* not only achieves the best (or sometimes second best) final fitting precision compared with other auxiliary objectives, it also dramatically accelerates the convergence with appearing a visibly larger slope in the loss curve. Therefore, *DTW* is still worthy to be applied despite it is time consuming.

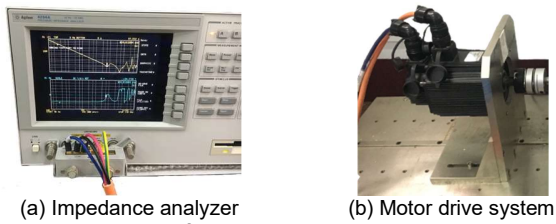
From the perspective of the best run (Fig. 11), FEMEIF still shows superior fitting precision than WBEIF. Fig. 9 visualizes the fitted impedance of the best run of the two methods under the same time consumption (WBEIF at 300th generation and FEMEIF at 146th generation), it can be seen that the FEMEIF achieves nearly the same impedance as the target impedance whereas the WBEIF exhibits obvious mismatch in several frequency ranges where a maximum of 12 dB fitting error occurs at 27 kHz.

Furthermore, optimization variance of WBEIF and FEMEIF (with all auxiliary objectives applied) under the same time cost is compared. Similar to the analysis of the mean fitting precision, the estimated sample variance in (10b) also exists a difference from the true population variance, which, leads to that directly compare S^2_{WSBGA} with S^2_{FEMGOA} unreliable. Thus, F-test, which is a kind of hypothesis testing technique, is applied to conduct a rigorous statistical comparison. The detailed procedures for using F-test to compare the variance of two series of data can be found in Ref. [32]. In short, if the ratio $S^2_{WSBGA} / S^2_{FEMGOA}$ is larger than a threshold which is determined by the F-distribution with a significant level α_F, it indicates that the true variance of WBEIF is significantly larger than the true variance of FEMEIF at a significance level of 1-α_F. Fig. 13(a) compares the variance of two algorithms under the same time cost. α_F is empirically chosen as 0.05 so the threshold is calculated as 1.394. It can be seen that since a very early 3.3min when both two algorithms are far from convergence, there is



(a) Case I (b) Case II

Fig. 13. Variance comparison between WBEIF and FEMEIF (all objectives applied) under the same time cost.



(a) Impedance analyzer (b) Motor drive system

Fig. 14. Test bench of the three-phase motor drive system.

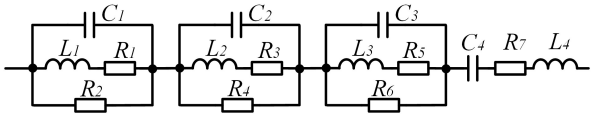


Fig. 15. Equivalent circuit used for fitting the common-mode

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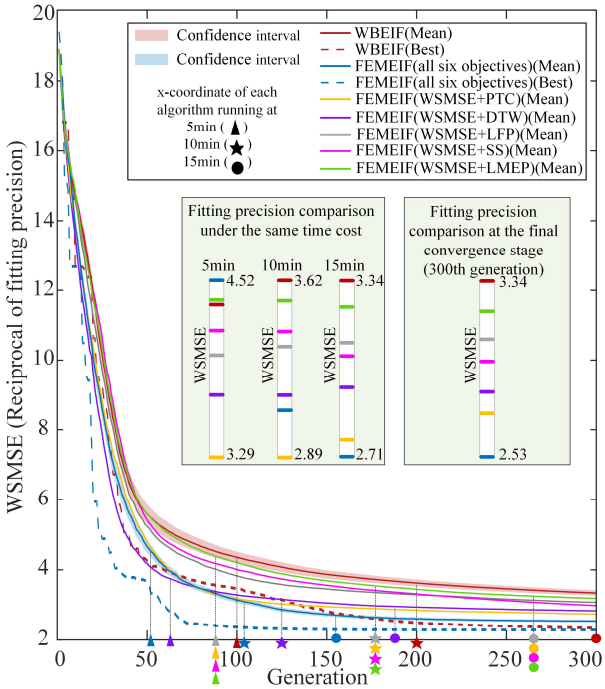


Fig. 16. Optimization process of WBEIF and FEMEIF (with different objective-combination) in Case II.

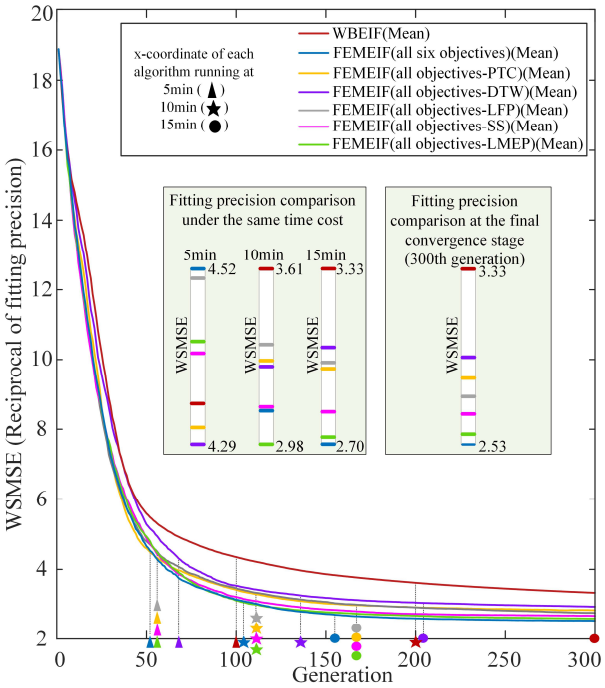


Fig. 17. Optimization process of WBEIF and FEMEIF (with different objective-combination) in Case II, "-" represents for "except for".

less than 5% risk of concluding that the true variance of FEMEIF starts to be lower than the true variance of WBEIF, which, verifies FEMEIF can get a stabler fitting performance owing to the appropriate guidance of the auxiliary objectives.

B. Case II: Motor drive system

For further verification, the proposed FEMEIF is applied in a real industrial application which is fitting the common-mode impedance of a motor drive system (including a 3-phase AC motor and cable) for EMI prediction and filter design [4]. Fig. 14 shows the test bench. Agilent 4294A is used to measure the common-mode impedance of the motor system. The equivalent circuit used for fitting is shown in Fig. 15.

Fig. 16 and Fig. 17 visualize the optimization process (averaged 100 runs and the best run) of the two optimization methods. Similar conclusions can be drawn as with Case I. Compared with WBEIF, FEMEIF with different objective combinations still achieves a much better fitting precision at the final convergence stage (300th generation). Each auxiliary objective can improve the fitting precision to different extent,

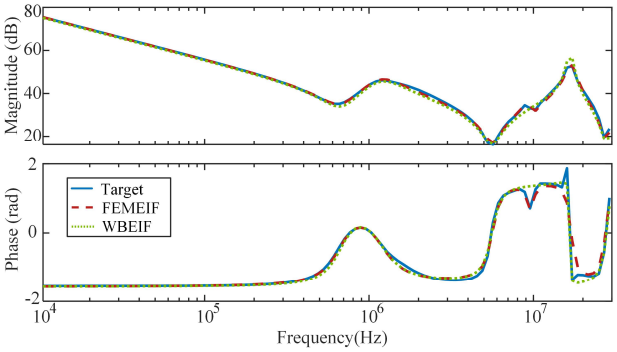


Fig. 18. Comparison of fitting results between WBEIF and FEMEIF (all objectives) at the best run under the same time cost (Case II).

and the combination of them brings the final fitting precision to the optimum. Under the same time consumption (5min, 10min, 15min), despite a higher computational complexity, FEMEIF still realizes a lower fitting error than WBEIF owing to the surplus compensation from the superior fitting precision.

When comparing the best run, the two methods finally reach nearly the same final fitting precision, this is because the utmost fitting capability of the selected equivalent circuit is easy to reach. But under the same time consumption (e. g. the FEMEIF at the 72nd generation of the best run and the WBEIF at the 140th generation of the best run), FEMEIF realizes a much-reduced *WSMSE* (FEMEIF is 2.46 and WBEIF is 2.95). Fig. 18 visualizes the fitted impedance of the FEMEIF's 72nd generation and WBEIF's 140th generation, the fitting error of WBEIF is much worse than FEMEIF with even omitting a peak and a trough at 8 MHz-10 MHz. Besides, according to the confidence interval, it can be deduced that FEMEIF can reach the utmost fitting capability of the equivalent circuit with a higher probability and with less rounds of tries. It is worth noting that the reason for that the fitting precision in Case II is not as good as Case I is because the limitation of the selected equivalent circuit, FEMEIF can be combined with a better-designed circuit to achieve a better fitting precision.

Moreover, Fig. 13(b) compares the variance in Case II, it can be seen that just after 5.9min where both two algorithms haven't converged, there is less than 5% risk of concluding that the true variance of FEMEIF is lower than the true variance of WBEIF, which, verifies that FEMEIF can still realize stabler fitting performance in this case.

In summary, although different combinations of the auxiliary objectives show different effectiveness in different practical applications, FEMEIF with any objective-combination can achieve superior fitting performance than the traditional

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WBEIF no matter compared under the same time cost or at the same generation. Particularly, although enabling multi-objective impedance fitting increases the time cost per generation, the use of multiple feature-based objectives can effectively help the optimization process of *WSMSE* escape from local minima, which, provides surplus compensation for the increased time cost and still can achieve better fitting precision under same time cost. Besides, since the feature-based auxiliary objectives can retain "shortcut" individuals from different perspectives, applying more objectives can achieve better fitting precision at the final convergence stage. However, it doesn't mean simultaneously applying all objectives is always the best choice when considering the time cost. For impedance fitting tasks where the fitting precision is the most critical thing or objective evaluation is expensive, all the auxiliary objectives are recommended to be applied. For time-critical applications, the applied objectives can be flexibly designed according to practical demand. We suggest to add objectives in the order of *PTC>SS≈LFP>DTW>LMEP* until a satisfactory tradeoff has been made between fitting precision and time cost. Alternatively, some light-cost many-objective algorithms can also be used to reduce the time cost.

In addition, the main drawback of the proposed FEMEIF is the hyperparameters increase with the applied objectives. To mitigate this problem, the proposed features which require a large number of hyperparameters (such as *SS*) or are less effective (such as *DTW*) can be removed. Besides, FEMEIF allows the exploration and application of other potentially effective feature engineering techniques which require less hyperparameter.

V. Conclusion

This paper proposes a FEMEIF for realizing automatic impedance fitting. FEMEIF utilizes several effective feature engineering techniques to fully leverage the limited impedance information, and then process these synergetic features into feature-based auxiliary objectives to enable a multi-objective impedance fitting. In this way, compared with the traditional WBEIF where only *WSMSE* is used as the single objective, effective "shortcut" individuals which are omitted by *WSMSE* could be retained and synergetically function during the optimization process of FEMEIF. As a result, it effectively helps the optimization process to escape from the local minimum. Experiments show that FEMEIF (with any objective-combination) can realize significantly better fitting precision (lower *WSMSE*) than the traditional WBEIF no matter comparing them at the final convergence stage or under the same time cost. Also, FEMEIF can realize a stabler fitting performance with reduced variance. Besides, the solutions of FEMEIF are also of higher comprehensive quality (better shape similarity, lower maximum error, etc.). FEMEIF requires no additional measurements or extra expenses. It is simple, flexible, and universal for equipping with any state-of-the-art many-objective optimization algorithms and equivalent circuits. Besides, FEMEIF provides an open topic for exploring other effective feature engineering to further improve the impedance fitting. More importantly, the concept of solving a single-objective task in a many-objective way with the "shortcut"-characterized feature engineering techniques can be generalized for improving other single-objective optimization applications in the industry.

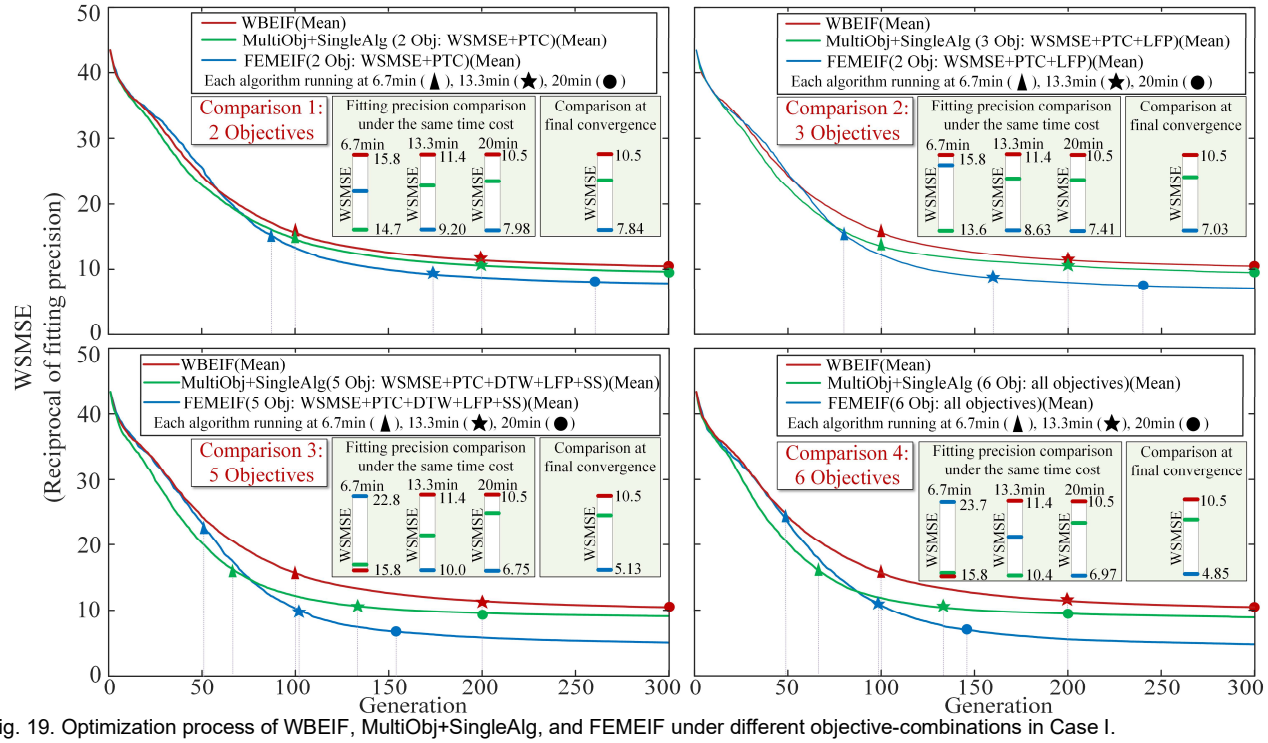


Fig. 19. Optimization process of WBEIF, MultiObj+SingleAlg, and FEMEIF under different objective-combinations in Case I.

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DISCUSSION

A. Necessity of using multi-/many-objective optimization algorithm for implementing FEMEIF

Since the proposed feature-based auxiliary objectives are not conflicting, the necessity of applying multi-/many-objective optimization algorithms is worth verifying due to the worry about less efficiency. Three algorithms are compared: the traditional WBEIF (single objective + single objective algorithm), FEMEIF (proposed multiple objectives + GrEA), and *MultiObj+SingleAlg* (proposed multiple objectives + single objective algorithm (configuration of the algorithm is same as WBEIF)). The objective of *MultiObj+SingleAlg* is

$$\begin{aligned}\text{Minimum } F(\boldsymbol{x}) = {} & lw_1 / F_{WSMSE} + lw_2 / F_{SS} + lw_3 / F_{PTC} \\ & + lw_4 / F_{LPF} + lw_5 / F_{DTW} + lw_6 / F_{LMEP}\end{aligned} \tag{11}$$

where lw_1-lw_6 denote the weight constants. lw_1-lw_6 are well-tuned until the algorithm performs the best. The experiments are verified on Case I and the results are shown in Fig. 19.

As can be seen, although *MultiObj+SingleAlg* realizes better fitting precision than WBEIF, which, further verifies the effectiveness of the proposed auxiliary objectives. However, it shows a severely deteriorated performance than the proposed FEMEIF either compared under the same time cost or at the final convergence stage. It is because many-objective algorithm can retain the "shortcut" individuals picked by auxiliary objectives in an appropriate distribution during the optimization process, which, enables them to constantly function to promote the optimization of *WSMSE*. Therefore, although it becomes less efficient of using multi-/many-objective algorithms to optimize multiple non-conflicting objectives, multi-/many-objective algorithms are still the best choice for combining multiple feature-based objectives.

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